Lecture 8: Infinite-Horizon Optimization and Dynamic Programming

Junnan Zhang

Paula and Gregory Chow Institute for Studies in Economics
Xiamen University

Fall, 2026

Chapter Overview

- Brief introduction to infinite-horizon optimization in discrete time under certainty
- Main purpose: introduce the theory and techniques of dynamic programming

The material is presented in four parts:

1. Introduces the problem and provides theoretical results necessary for applications of stationary dynamic programming techniques
2. Additional mathematical tools
3. Applications

Outline

The canonical discrete-time infinite-horizon optimization program:

$$\sup_{\{x_t, y_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \tilde{U}(t, x_t, y_t)$$

$$\begin{aligned} \text{s.t. } & y_t \in \tilde{G}(t, x_t) \text{ for all } t \geq 0 \\ & x_{t+1} = \tilde{f}(t, x_t, y_t) \text{ for all } t \geq 0 \\ & x_0 \text{ given} \end{aligned}$$

- $\beta \in [0, 1)$ is the discount factor
- $x_t \in X \subset \mathbb{R}^{K_x}$ and $y_t \in Y \subset \mathbb{R}^{K_y}$ for some $K_x, K_y \geq 1$
- x_t: state variables (state vector)
- y_t: control variables (control vector)
- $\tilde{U} : \mathbb{Z}_+ \times X \times Y \to \mathbb{R}$ is the instantaneous payoff function
- $\tilde{G}(t, x)$ is a set-valued mapping (correspondence): $\tilde{G} : \mathbb{Z}_+ \times X \rightrightarrows Y$

Constraints and Evolution

- First constraint: $y_t \in \tilde{G}(t, x_t)$ specifies what values of control vector y_t are allowed, given state x_t at time t
- Evolution equation: $\tilde{f} : \mathbb{Z}_+ \times X \times Y \to X$ specifies evolution of state vector as function of last period's state and control vectors
- This formulation highlights distinction between state and control variables
- Often more convenient to work with transformation to eliminate y_t

Problem 6.1: Transformed Formulation

$$V^*(0, x_0) = \sup_{\{x_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(t, x_t, x_{t+1})$$

$$\text{s.t. } x_{t+1} \in G(t, x_t) \text{ for all } t \geq 0$$

$$x_0 \text{ given}$$

- $x_t \in X \subset \mathbb{R}^K$ (now $K = K_x$)
- x_t corresponds to state vector, x_{t+1} plays role of control vector at time t
- $U : \mathbb{Z}_+ \times X \times X \to \mathbb{R}$ is instantaneous payoff function
- $G : \mathbb{Z}_+ \times X \rightrightarrows X$ specifies constraint correspondence

- $V^* : \mathbb{Z}_+ \times X \to \mathbb{R}$ is the **value function**, which specifies supremum (highest possible value) that objective function can reach starting with some x_t at time t
- When maximal value is attained by sequence $\{x_{t+1}^*\}_{t=0}^{\infty} \in X^{\infty}$, we refer to this as a **solution** or an **optimal plan**
- Objective: characterize optimal plan $\{x_{t+1}^*\}_{t=0}^{\infty}$ and value function $V^*(0, x_0)$
- Here, X^{∞} is countable infinite product of set X. If X is bounded, then $X^{\infty} \subset \ell^{\infty}$, where ℓ^{∞} is the vector space of infinite sequences that are bounded with the sup norm $\|\cdot\|_{\infty}$ (or simply $\|\cdot\|$)

Consider an optimal growth problem:

$$\max_{\{k_t, c_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

$$\text{s.t. } k_{t+1} = f(k_t) + (1-\delta)k_t - c_t$$

$$k_t \geq 0, \text{ given } k_0 > 0, \text{ and } u : \mathbb{R}_+ \to \mathbb{R}$$

Let $x_t = k_t$ and $x_{t+1} = k_{t+1}$. The objective function can be written as:

$$\max_{\{k_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(f(k_t) - k_{t+1} + (1-\delta)k_t)$$

subject to $k_t \geq 0$. This is special case of Problem 6.1 with:

- $U(t, k_t, k_{t+1}) = u(f(k_t) - k_{t+1} + (1-\delta)k_t)$
- Constraint correspondence $G(t, k_t) = [0, f(k_t) + (1-\delta)k_t]$ (since $c_t \geq 0$)

Stationarity Feature

- Notable feature emphasized by Example 6.1: U and G do not explicitly depend on time
- This feature is fairly common in economics and many interesting problems can be formulated in **stationary form**
- Stationary problem involves:
  - Objective function that is a discounted sum
  - U and G functions that do not explicitly depend on time

Outline

Consider the stationary form of Problem 6.1.

Problem 6.2:

$$V^*(x_0) = \sup_{\{x_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(x_t, x_{t+1})$$
$$\text{s.t. } x_{t+1} \in G(x_t) \text{ for all } t \geq 0$$
$$x_0 \text{ given}$$

- Constraint correspondence: $G : X \rightrightarrows X$
- Instantaneous payoff functions: $U : X \times X \to \mathbb{R}$
- Value function without time argument: $V^*(x_0)$

- Problem 6.2 corresponds to the **sequence problem**: involves choosing infinite sequence $\{x_t\}_{t=0}^{\infty} \in X^{\infty}$
- Sequence problems sometimes have nice features, but solutions often difficult to characterize analytically and numerically
- Basic idea of dynamic programming: turn sequence problem into **functional equation**
- Transform problem into one of finding a function rather than a sequence

Problem 6.3:

$$V(x) = \sup_{y \in G(x)} \{U(x, y) + \beta V(y)\}, \quad \text{for all } x \in X$$

where $V : X \to \mathbb{R}$.

- This functional equation is also called the **"Bellman equation"** after Richard Bellman
- Instead of explicitly choosing a sequence $\{x_t\}_{t=0}^{\infty}$, we choose a **policy** $y \in G(x) \subset X$ for given $x \in X$
- Since $U(\cdot, \cdot)$ does not depend on time, no reason for policy to be time-dependent either
- Use V for function defined in Problem 6.3 and V^* for function defined in Problem 6.2

Economic Interpretation

$$V(x) = \sup_{y \in G(x)} \{U(x, y) + \beta V(y)\}, \quad \text{for all } x \in X$$

- This is a **"recursive formulation"**: function $V(x)$ appears on both left- and right-hand sides, defined recursively
- In many cases, solution to Problem 6.3 is simpler to characterize analytically than the corresponding solution to sequence problem
- Question: when are the two problems equivalent?

Natural Derivation of the Bellman Equation

Suppose Problem 6.2 has maximum starting at x_0 attained by optimal sequence $\{x_t^*\}_{t=0}^{\infty}$ with $x_0^* = x_0$. Under weak technical conditions:

$$\begin{aligned} V^*(x_0) &= \sum_{t=0}^{\infty} \beta^t U(x_t^*, x_{t+1}^*) \\ &= U(x_0, x_1^*) + \beta \sum_{s=0}^{\infty} \beta^s U(x_{s+1}^*, x_{s+2}^*) \\ &= U(x_0, x_1^*) + \beta V^*(x_1^*) \end{aligned}$$

This encapsulates the basic idea of dynamic programming: **Principle of Optimality**, which states that optimal plan can be broken into two parts:

1. What is optimal today
2. The optimal continuation path

Optimal Policy

$$V(x) = \sup_{y \in G(x)} \{U(x, y) + \beta V(y)\}, \quad \text{for all } x \in X$$

- Solution can be represented by a time invariant **policy function** (or policy mapping) $\pi : X \to X$ that determines which value of x_{t+1} to choose for given state variable x_t
- Two complications:
  1. Control reaching optimal value may not exist (reason for "sup" notation)
  2. Solution may involve policy correspondence $\Pi : X \rightrightarrows X$ (more than one maximizer)
- Ignore these complications for now

Characterizing Policy Function

Once value function V is determined, policy function is straightforward to characterize.
If the optimal policy is given by $\pi(x)$, then:

$$V(x) = U(x, \pi(x)) + \beta V(\pi(x)) \quad \text{for all } x \in X$$

- This equation follows from the fact that $\pi(x)$ is the optimal policy
- When $y = \pi(x)$, the right-hand side of the Bellman equation reaches maximal value $V(x)$
- This is one way of determining the policy function

Usefulness of Recursive Formulation

- There are powerful tools that establish existence of solution and some of its properties
- Next section: present results on relationship between solution to the sequence problem (Problem 6.2) and the recursive formulation (Problem 6.3)
- Establish results on concavity, monotonicity, and differentiability of the value function

Outline

1. Problem Description
2. Stationary Dynamic Programming
3. Stationary Dynamic Programming Results
4. The Contraction Mapping Theorem
5. Proofs of Main Theorems
6. Applications of Stationary Dynamic Programming
   - Euler Equation
   - Dynamic Programming vs Sequence Problem

Main purpose:

- Ensure sequence $\{x_t^*\}_{t=0}^{\infty} \in X^{\infty}$ that attains supremum in Problem 6.2 satisfies recursive equation of dynamic programming

$$V(x_t^*) = U(x_t^*, x_{t+1}^*) + \beta V(x_{t+1}^*), \quad \text{for all } t = 0, 1, \ldots$$

- The solution to the Bellman equation is also a solution to Problem 6.2
- In other words, establish equivalence results between solutions to Problems 6.2 and 6.3

Feasible Sequences

Define set of feasible sequences or plans starting with initial value x_t:

$$\Phi(x_t) = \{\{x_s\}_{s=t}^{\infty} : x_{s+1} \in G(x_s) \text{ for } s = t, t+1, \dots\}$$

- Intuitively, $\Phi(x_t)$ is the set of feasible choices of vectors starting from x_t
- Denote typical element of $\Phi(x_0)$ by $\mathbf{x} = (x_0, x_1, \dots) \in \Phi(x_0)$
- Want sequence to satisfy:

$$V(x_t^*) = U(x_t^*, x_{t+1}^*) + \beta V(x_{t+1}^*) \text{ for all } t = 0, 1, \dots$$

Assumption 6.1

$G(x)$ is nonempty for all $x \in X$; and for all $x_0 \in X$ and $\mathbf{x} \in \Phi(x_0)$,

$$\lim_{n \to \infty} \sum_{t=0}^{n} \beta^t U(x_t, x_{t+1}) \text{ exists and is finite}$$

- Assumption stronger than necessary for theory: for much of dynamic programming theory, it is sufficient that the limit exists
- But in economic applications, we are not interested in optimization problems where agents achieve infinite value

Theorem 6.1 (Equivalence of Values)

Suppose Assumption 6.1 holds. Then for any $x \in X$, any solution $V^*(x)$ to Problem 6.2 is also a solution to Problem 6.3. Moreover, any solution $V(x)$ to Problem 6.3 is also a solution to Problem 6.2, so that $V^*(x) = V(x)$ for all $x \in X$.

- Under Assumption 6.1, both sequence problem and recursive formulation achieve the same value
- Fundamental result that justifies the dynamic programming approach

Theorem 6.2 (Principle of Optimality)

Suppose Assumption 6.1 holds. Let $\mathbf{x}^* \in \Phi(x_0)$ be a feasible plan that attains $V^*(x_0)$ in Problem 6.2. Then

$$V^*(x_t^*) = U(x_t^*, x_{t+1}^*) + \beta V^*(x_{t+1}^*) \tag{6.3}$$

for $t = 0, 1, \ldots$, with $x_0^* = x_0$.
Moreover, if any $\mathbf{x}^* \in \Phi(x_0)$ satisfies equation (6.3), then it attains the optimal value in Problem 6.2.

- If any feasible plan satisfies the Bellman equation, then it attains optimal value
- We can go from solution of the Bellman equation to solution of the sequence problem and vice versa

Example 6.4: Consider an optimal growth model with log preferences, Cobb-Douglas technology, and full depreciation:

$$\max_{\{k_t, c_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \log c_t$$

$$\text{s.t. } k_{t+1} = k_t^{\alpha} - c_t$$

$$k_0 > 0$$

where $\beta \in (0, 1)$, k is capital-labor ratio, and the resource constraint follows from the production function $K^{\alpha} L^{1-\alpha}$ in per capita terms.

Closed-Form Solution

We solve the Bellman equation

$$V(x) = \max_{y \geq 0} \{\log(x^\alpha - y) + \beta V(y)\}.$$

Standard approach for log–Cobb–Douglas:

- Conjecture that the value function is logarithmic:

$$V(x) = A + B \log x.$$

- Use the Bellman equation to solve for unknown constants (A, B).
- Obtain closed-form policy and value functions.

- Guess:
$$V(x) = A + B \log x.$$
- Bellman equation:
$$A + B \log x = \max_y \left\{ \log(x^\alpha - y) + \beta \left[A + B \log y \right] \right\}.$$
- FOC w.r.t. y:
$$-\frac{1}{x^\alpha - y} + \frac{\beta B}{y} = 0 \implies y = \frac{\beta B}{1 + \beta B} x^\alpha.$$
  Using $B = \frac{\alpha}{1-\alpha\beta}$ gives $y = \alpha\beta x^\alpha$.
- Optimal next-period capital:
$$k_{t+1} = y^*(x) = \alpha\beta x^\alpha.$$

- Substitute $y^*(x) = \alpha\beta x^\alpha$ into Bellman equation:

$$\begin{aligned} A + B\log x &= \log\left((1-\alpha\beta)x^\alpha\right) + \beta\left[A + B\log(\alpha\beta x^\alpha)\right] \\ &= \log(1-\alpha\beta) + \alpha\log x + \beta A + \beta B\left(\log(\alpha\beta) + \alpha\log x\right). \end{aligned}$$

- Match coefficients on $\log x$:

$$B = \alpha + \beta B\alpha \implies B = \frac{\alpha}{1-\alpha\beta}$$

- Match constant terms:

$$A = \log(1-\alpha\beta) + \beta A + \beta B\log(\alpha\beta).$$

  Solve:

$$A = \frac{1}{1-\beta}\left[\log(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta}\log(\alpha\beta)\right].$$

Continuity and Compactness Assumptions

Assumption 6.2

X is a compact subset of $\mathbb{R}^K$, G is nonempty-valued, compact-valued, and continuous. Moreover, $U : X_G \to \mathbb{R}$ is continuous, where $X_G = \{(x, y) \in X \times X : y \in G(x)\}$.

- Most restrictive: the state space is compact
- Since X is compact and $G(x)$ is compact-valued, X_G is also compact
- Continuous function on compact domain: U is bounded

Theorem 6.3 (Existence of Solutions)

Suppose that Assumptions 6.1 and 6.2 hold. Then there exists a unique continuous and bounded function $V : X \to \mathbb{R}$ that satisfies the Bellman equation. Moreover, for any $x_0 \in X$, an optimal plan $\mathbf{x}^* \in \Phi(x_0)$ exists.

Two major results:

1. **Existence and uniqueness** of the value function
2. **Existence** of an optimal plan

Combined with Theorem 6.1: optimal policy function achieving supremum V^* in Problem 6.2 exists and V^* is continuous and bounded.

Non-uniqueness of Optimal Plans

- Optimal plan in Problem 6.2 (or 6.3) may not be unique, even though the value function is unique: for example, when two alternative feasible sequences achieve same maximal value
- As in static optimization problems, non-uniqueness because of the lack of strict concavity of objective function
- When we include a concavity assumption, uniqueness of optimal plan is guaranteed

Concavity Assumption

Assumption 6.3

U is concave; that is, for any $\alpha \in (0,1)$ and any $(x,y),(x',y') \in X_G$:

$$U(\alpha x + (1-\alpha)x', \alpha y + (1-\alpha)y') \geq \alpha U(x,y) + (1-\alpha)U(x',y')$$

and the inequality is strict if $x \neq x'$.
In addition, G is convex in the sense that for any $\alpha \in [0,1]$, and $x, x', y, y' \in X$ such that $y \in G(x)$ and $y' \in G(x')$:

$$\alpha y + (1-\alpha)y' \in G(\alpha x + (1-\alpha)x').$$

Concavity of Value Function

Theorem 6.4 (Concavity of the Value Function)

Suppose that Assumptions 6.1, 6.2, and 6.3 hold. Then the unique $V : X \to \mathbb{R}$ that satisfies the Bellman equation is strictly concave.

- If Assumption 6.3 relaxed so that U is concave (without additional requirement for $x \neq x'$), then a weaker version applies and implies V is concave
- Important for uniqueness of optimal plans

Unique Optimal Plan and Policy Function

Corollary 6.1

Suppose Assumptions 6.1, 6.2, and 6.3 hold. Then there exists a unique optimal plan $x^* \in \Phi(x_0)$ for any $x_0 \in X$. Moreover, the optimal plan can be expressed as:

$$x_{t+1}^* = \pi(x_t^*)$$

where $\pi : X \to X$ is a continuous policy function.

- The policy correspondence becomes a function, not just a correspondence.
- The policy function is continuous.

Monotonicity Assumption

Assumption 6.4

For each $y \in X$, $U(\cdot, y)$ is strictly increasing in each of its first K arguments, and G is monotone in the sense that $x \leq x'$ implies $G(x) \subset G(x')$.

- Ensures payoff function increasing in state variables
- Larger values of state variables attractive from viewpoint of relaxing constraints
- Natural in economic applications where "more is better"

Theorem 6.5 (Monotonicity)

Suppose Assumptions 6.1, 6.2, and 6.4 hold, and let $V : X \to \mathbb{R}$ be the unique solution to the Bellman equation. Then V is strictly increasing in all of its arguments.

- Consistent with monotonic preferences
- Useful for comparative statics analysis
- In growth models, higher capital stock always increases value function

Differentiability Assumption

Assumption 6.5

U is continuously differentiable on the interior of its domain X_G.

- Common in most economic models
- Enables us to work with first-order necessary conditions

Theorem 6.6 (Differentiability)

Suppose Assumptions 6.1, 6.2, 6.3, and 6.5 hold. Let $\pi(\cdot)$ be the policy function from Corollary 6.1, and assume $x \in \text{Int } X$ and $\pi(x) \in \text{Int } G(x)$. Then $V(\cdot)$ is differentiable at x, with gradient:

$$DV(x) = D_x U(x, \pi(x))$$

- Allows us to use differential calculus on the Bellman equation
- Enables derivation of Euler equations

Summary

Given these assumptions, the following results are established:

1. **Theorem 6.1:** Equivalence of Values
2. **Theorem 6.2:** Principle of Optimality
3. **Theorem 6.3:** Existence of Solutions
4. **Theorem 6.4:** Concavity of Value Function
5. **Theorem 6.5:** Monotonicity of Value Function
6. **Theorem 6.6:** Differentiability of Value Function and the Envelope Theorem

Outline

Metric Space: (S, d) where S is a space and d is a distance metric
Operators: Mappings $T : S \to S$

Definition 6.1 (Contraction Mapping)

Let (S, d) be a metric space and $T : S \to S$ be an operator. If for some $\beta \in (0, 1)$:

$$d(Tz_1, Tz_2) \leq \beta d(z_1, z_2) \text{ for all } z_1, z_2 \in S$$

then T is a contraction mapping (with modulus β).

Intuition: Contraction mappings bring elements uniformly closer together.

Example 6.2: On interval $S = [a, b]$ with usual metric $d(z_1, z_2) = |z_1 - z_2|$: T is a contraction if:

$$\frac{|Tz_1 - Tz_2|}{|z_1 - z_2|} \leq \beta < 1 \text{ for all } z_1 \neq z_2$$

This means T has slope less than 1 everywhere.

Definition 6.2 (Fixed Point)

A fixed point of T is any element $\hat{z} \in S$ satisfying $T\hat{z} = \hat{z}$.

Theorem 6.7 (Contraction Mapping Theorem)

Let (S, d) be a complete metric space and suppose $T : S \to S$ is a contraction. Then T has a unique fixed point $\hat{z}$; that is, there exists a unique $\hat{z} \in S$ such that:

$$T\hat{z} = \hat{z}.$$

Moreover, $T^n z \to \hat{z}$ as $n \to \infty$ for any $z \in S$.

- Simple conditions that guarantee existence, uniqueness, and convergence
- Applies to infinite-dimensional function spaces
- Foundation of dynamic programming theory

Theorem 6.8

Let (S, d) be complete and $T : S \to S$ be a contraction with fixed point $\hat{z}$.

1. If S' is a closed subset of S and $T(S') \subseteq S'$, then $\hat{z} \in S'$
2. If $T(S') \subseteq S'' \subseteq S'$, then $\hat{z} \in S''$

- If we start in a closed set that maps to itself, fixed point is in that set
- Enables proving properties like strict concavity or monotonicity

Checking contraction property directly is often difficult. Blackwell provides easier conditions:

Theorem 6.9 (Blackwell's Conditions)

Let $X \subseteq \mathbb{R}^K$ and $B(X)$ be the space of bounded functions $f : X \to \mathbb{R}$ equipped with the sup norm. If $T : B(X) \to B(X)$ satisfies:

1. Monotonicity: For any $f, g \in B(X)$, $f(x) \leq g(x)$ for all x implies $(Tf)(x) \leq (Tg)(x)$ for all $x \in X$; and
2. Discounting: $\exists \beta \in (0, 1)$ such that

$$[T(f + c)](x) \leq (Tf)(x) + \beta c$$

   for all $f \in B(X)$, $c \geq 0$, and $x \in X$.

Then T is a contraction with modulus β on $B(X)$.

Applying Blackwell's Conditions to Dynamic Programming

For the Bellman operator:

$$TV(x) = \max_{y \in G(x)} \{U(x, y) + \beta V(y)\}$$

- Checking Monotonicity: If $V_1(x) \leq V_2(x)$ for all x, then:

$$TV_1(x) = \max_{y \in G(x)} \{U(x, y) + \beta V_1(y)\} \leq \max_{y \in G(x)} \{U(x, y) + \beta V_2(y)\} = TV_2(x)$$

- Checking Discounting:

$$\begin{aligned} T(V + c)(x) &= \max_{y \in G(x)} \{U(x, y) + \beta(V(y) + c)\} \\ &= \max_{y \in G(x)} \{U(x, y) + \beta V(y)\} + \beta c \\ &= TV(x) + \beta c \end{aligned}$$

Outline

For a feasible infinite sequence $x = (x_0, x_1, \ldots) \in \Phi(x_0)$, define

$$\bar{U}(x) \equiv \sum_{t=0}^{\infty} \beta^t U(x_t, x_{t+1})$$

Lemma 6.1

Suppose Assumption 6.1 holds. For any $x_0 \in X$ and $x \in \Phi(x_0)$:

$$\bar{U}(x) = U(x_0, x_1) + \beta \bar{U}(x')$$

where $x' = (x_1, x_2, \ldots)$.

Proof Sketch: Theorem 6.1 (Equivalence)

Goal: Show $V^*(x) = V(x)$ where:

- $V^*(x) = \sup_{z \in \Phi(x)} \bar{U}(\mathbf{z})$ (Problem 6.2)
- $V(x) = \sup_{y \in G(x)} \{U(x, y) + \beta V(y)\}$ (Problem 6.3)

Idea: Use Lemma 6.1 and the definition of supremum. (Read pp. 195–197 of the textbook.)

Proof Sketch: Theorem 6.2 (Principle of Optimality)

- Let x^* attain $V^*(x_0)$ in Problem 6.2. Our goal is to show that

$$V^*(x_t^*) = U(x_t^*, x_{t+1}^*) + \beta V^*(x_{t+1}^*)$$

  Proof idea: show that $(x_t^*, x_{t+1}^*, \ldots)$ is optimal starting from x_t^* using Lemma 6.1 and induction
- Suppose x^* satisfies the Bellman equation. We want to show that it is optimal.
  Proof idea: iterating on the Bellman equation.

Two Proofs of Theorem 6.3 (Existence)

Version 1 (Sequence problem under compact X):

- Basic idea: show that the objective function is continuous on a compact set, and therefore attains a maximum
- Treat the objective function as continuous on X^{∞}
- Show that the constraint set is a compact subset of X^{∞}

Version 2 (Contraction Mapping):

- Define Bellman operator T on space $C(X)$ of continuous functions
- Show T satisfies Blackwell's conditions
- Apply Contraction Mapping Theorem: unique fixed point exists
- Fixed point is solution to recursive problem

Outline

Applications

- We show how dynamic programming can be applied to a range of problems
- **Main Result**: Theorem 6.10 shows how dynamic first-order conditions (Euler equations) together with the transversality condition are sufficient to characterize solutions

Outline

The Functional Equation

Consider the functional equation corresponding to Problem 6.3:

$$V(x) = \max_{y \in G(x)} \{U(x, y) + \beta V(y)\} \quad \text{for all } x \in X$$

Throughout, we assume Assumptions 6.1–6.5 hold.

- From Theorem 6.4: maximization problem is strictly concave
- From Theorem 6.6: maximand is differentiable
- For interior solutions $y \in \text{Int } G(x)$: first-order conditions are necessary and sufficient

For optimal solutions, we can characterize them by the **Euler equations**:

$$D_y U(x, y^*) + \beta DV(y^*) = 0$$

- Use the Envelope Theorem for dynamic programming by differentiating the functional equation with respect to state vector x:
$$DV(x) = D_x U(x, y^*)$$
- Substituting it into the Euler equation gives

$$D_y U(x, \pi(x)) + \beta D_x U(\pi(x), \pi(\pi(x))) = 0$$

This is a **functional equation** in the unknown function $\pi(.)$ and characterizes the optimal policy function.

One-Dimensional Case

When both x and y are real numbers, the Euler equation becomes:

$$\frac{\partial U(x, y^*)}{\partial y} + \beta V'(y^*) = 0$$

Sum of marginal gain today from increasing y plus discounted marginal gain from increasing y on future returns must equal zero

The one-dimensional Envelope Condition gives:

$$V'(x) = \frac{\partial U(x, y^*)}{\partial x}$$

Combining with the Euler equation:

$$\frac{\partial U(x, \pi(x))}{\partial y} + \beta \frac{\partial U(\pi(x), \pi(\pi(x)))}{\partial x} = 0$$

With time arguments explicitly:

$$\frac{\partial U(x_t, x^*_{t+1})}{\partial y} + \beta \frac{\partial U(x^*_{t+1}, x^*_{t+2})}{\partial x} = 0$$

- Euler equation is **not sufficient** for optimality. We also need the transversality condition.
- The transversality condition ensures that there are no beneficial simultaneous changes in an infinite number of choice variables.
- **General case**:

$$\lim_{t\to\infty} \beta^t D_x U(x_t^*, x_{t+1}^*) \cdot x_t^* = 0$$

- **One-dimensional case**:

$$\lim_{t\to\infty} \beta^t \frac{\partial U(x_t^*, x_{t+1}^*)}{\partial x} \cdot x_t^* = 0$$

- The transversality condition requires that the marginal return from state variable x times the value of this state variable does not increase asymptotically at rate faster than or equal to $1/\beta$.

Theorem 6.10 (Euler Equations and Transversality Condition)

Let $X \subset \mathbb{R}^K_+$ and suppose Assumptions 6.1–6.5 hold. Then a sequence $\{x_t^*\}_{t=0}^{\infty}$ such that $x_{t+1}^* \in \operatorname{Int} G(x_t^*)$, $t = 0, 1, \ldots$, is optimal for Problem 6.2 given x_0 if and only if it satisfies:

$$D_y U(x_t^*, x_{t+1}^*) + \beta D_x U(x_{t+1}^*, x_{t+2}^*) = 0$$

$$\lim_{t \to \infty} \beta^t D_x U(x_t^*, x_{t+1}^*) \cdot x_t^* = 0$$

The Euler equations together with the transversality condition are **necessary and sufficient**, under those assumptions.

Example 6.4: Consider an optimal growth model with log preferences, Cobb-Douglas technology, and full depreciation:

$$\max_{\{k_t, c_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \log c_t$$

$$\text{s.t. } k_{t+1} = k_t^{\alpha} - c_t$$

$$k_0 > 0$$

where $\beta \in (0,1)$, k is capital-labor ratio, and the resource constraint follows from the production function $K^{\alpha} L^{1-\alpha}$ in per capita terms.

Write down the Bellman equation

$$V(x) = \max_{y \geq 0} \{\log(x^{\alpha} - y) + \beta V(y)\}$$

where x is today's capital stock and y is tomorrow's capital stock.

- Previously, we used guess and verify to solve the value function.
- After a suitable compact restriction, Assumptions 6.1–6.5 hold, so Theorems 6.1–6.6 apply and we can use the Euler equation.
- Technical step: For fixed $k_0 > 0$, impose $c_t, k_t \geq \epsilon > 0$ with ϵ sufficiently small, then use concavity to verify optimality of the uniformly interior candidate after removing these bounds.

Applying Euler Equations

- Since $V(.)$ is differentiable, the first-order condition is

$$\frac{1}{x^{\alpha} - y} = \beta V'(y)$$

- Envelope Condition:

$$V'(x) = \frac{\alpha x^{\alpha-1}}{x^{\alpha} - y}$$

- Combining using $y = \pi(x)$:

$$\frac{1}{x^{\alpha} - \pi(x)} = \beta \frac{\alpha \pi(x)^{\alpha-1}}{\pi(x)^{\alpha} - \pi(\pi(x))}$$

This is a functional equation in a single function $\pi(x)$.

Guess-and-Verify Solution

- There are no straightforward ways to solve functional equations, but guess-and-verify often works.
- Let us conjecture that $\pi(x) = ax^{\alpha}$
- Substituting into the Euler equation:

$$\frac{1}{x^{\alpha} - ax^{\alpha}} = \beta \frac{\alpha a^{\alpha-1} x^{\alpha(\alpha-1)}}{a^{\alpha} x^{\alpha^2} - a^{1+\alpha} x^{\alpha^2}} = \frac{\beta}{a} \frac{\alpha}{x^{\alpha} - ax^{\alpha}}$$

which implies $a = \beta\alpha$ satisfies the equation.

Complete Solution

- The policy function is: $\pi(x) = \beta\alpha x^\alpha$
- Law of motion: $k_{t+1} = \beta\alpha k_t^\alpha$
- Optimal consumption: $c_t = (1-\beta\alpha)k_t^\alpha$
- Capital-labor ratio k_t converges to steady state k^*, which ensures transversality condition
- By Corollary 6.1 and Theorem 6.10: $\pi(x) = \beta\alpha x^\alpha$ is the unique policy function

Example 6.5: Consider an infinitely-lived consumer who solves

$$\max_{\{c_t, a_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

$$\text{subject to: } a_{t+1} = (1+r)a_t + w - c_t, \quad c_t \geq 0$$

- $u(c)$ is strictly increasing, continuously differentiable, strictly concave
- Without further constraints, this is not well-defined and allows consumer to build unlimited debt ($\lim_{t\to\infty} a_t = -\infty$)—"Ponzi games".

Three approaches to prevent unlimited borrowing:

1. **No-Ponzi condition**: Rule out such schemes directly
2. **No borrowing**: $a_{t+1} \geq 0$ for all t (too restrictive)
3. **Natural debt limit**: Maximum repayable debt $a_{t+1} \geq \underline{a} \equiv -w/r$

Ensuring Compactness

- **Challenge**: Assets a don't necessarily belong to a compact set.
- **Solution**: Choose upper bound $\bar{a}$ and restrict $a \in [\underline{a}, \bar{a}]$.
  1. Solve problem on a compact set
  2. Verify that the solution indeed lies in the state space
- **Natural choice**: $\bar{a} \equiv a_0 + w/r < \infty$

Recursive Formulation

- State variable: a_t
- Consumption: $c_t = (1 + r)a_t + w - a_{t+1} \geq 0$
- The Bellman equation

$$V(a) = \max_{a' \in [\underline{a}, \bar{a}],\, a' \leq (1+r)a+w} \{u((1 + r)a + w - a') + \beta V(a')\}$$

Verify that the assumptions are satisfied.

- Using one-dimensional Euler equation and the Envelope Condition gives

$$u'(c) = \beta(1+r)u'(c')$$

- Since $u'(.)$ exists and is strictly decreasing (because u is continuously differentiable and strictly concave), we get the following simple rule:

$$\text{if } r = \beta^{-1} - 1, c = c' \text{ (constant consumption)}$$
$$\text{if } r > \beta^{-1} - 1, c < c' \text{ (increasing consumption)}$$
$$\text{if } r < \beta^{-1} - 1, c > c' \text{ (decreasing consumption)}$$

Outline

Consider a dynamic optimization problem with finite horizon T:

$$\max_{\{x_t\}_{t=1}^{T+1}} \sum_{t=0}^{T} \beta^t U(x_t, x_{t+1})$$
$$\text{s.t. } x_{t+1} \geq 0, \; x_0 \text{ given}$$

- When $0 \leq t \leq T-1$, FOCs from finite-dimensional optimization assuming interior solution:

$$\frac{\partial U(x_t^*, x_{t+1}^*)}{\partial y} + \beta \frac{\partial U(x_{t+1}^*, x_{t+2}^*)}{\partial x} = 0$$

- When $t = T$, the complementary slackness condition implies:

$$x_{T+1}^* \geq 0, \text{ and } \beta^T \frac{\partial U(x_T^*, x_{T+1}^*)}{\partial y} x_{T+1}^* = 0$$

Example 6.6: In the optimal growth problem,

$$U(x_t, x_{t+1}) = u(f(x_t) + (1-\delta)x_t - x_{t+1})$$

Suppose that the world ends at date T. Then at the last date T:

$$\frac{\partial U(x_T^*, x_{T+1}^*)}{\partial y} = -u'(c_T^*) < 0$$

From the boundary condition, an optimal path must have $k_{T+1}^* = x_{T+1}^* = 0$.
No capital should be left at end of world. Any leftover resources should be consumed.

- The transversality condition can be derived heuristically by taking the limit of the boundary condition

$$\lim_{T\to\infty} \beta^T \frac{\partial U(x_T^*, x_{T+1}^*)}{\partial y} x_{T+1}^* = 0$$

- The Euler equation implies

$$\frac{\partial U(x_T^*, x_{T+1}^*)}{\partial y} + \beta \frac{\partial U(x_{T+1}^*, x_{T+2}^*)}{\partial x} = 0$$

- Substituting and changing timing:

$$\lim_{T\to\infty} \beta^T \frac{\partial U(x_T^*, x_{T+1}^*)}{\partial x} x_T^* = 0$$

which is exactly the transversality condition: a "boundary condition at infinity."